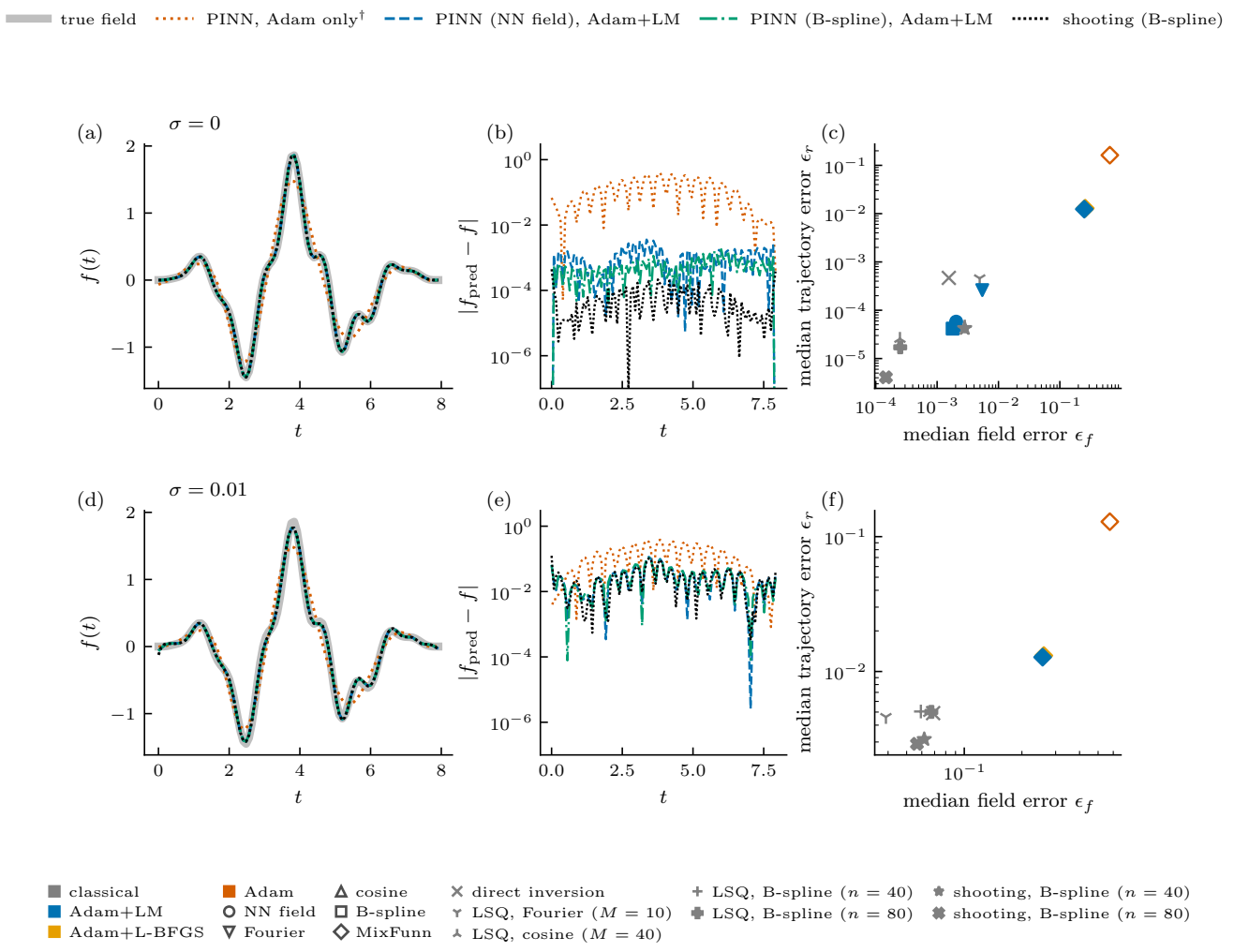

true field
PINN, Adam only†
PINN (NN field), Adam+LM
PINN (B-spline), Adam+LM
shooting (B-spline)
(a)
σ = 0
f(t)
t
(b)
|f_pred − f|
t
(c)
median trajectory error ϵ_r
median field error ϵ_f
(d)
σ = 0.01
f(t)
t
(e)
|f_pred − f|
t
(f)
median trajectory error ϵ_r
median field error ϵ_f
classical
Adam+LM
Adam+L-BFGS
Adam
NN field
Fourier
cosine
B-spline
MixFunn
direct inversion
LSQ, Fourier (M = 10)
LSQ, cosine (M = 40)
LSQ, B-spline (n = 40)
LSQ, B-spline (n = 80)
shooting, B-spline (n = 40)
shooting, B-spline (n = 80)